Onyx — ImageKit reference

Onyx · multi-process OS on Raspberry Pi 4

*The reference of **ImageKit** (`SD:/lib/imagekit.so`, `user/Kits/imagekit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: The Kits.*

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What it is

ImageKit is the system's one picture codec: BMP, GIF, PNG, JPEG, PCX and WebP read; PNG, JPEG, BMP and GIF written; resizing, turning, the photo adjustments.

Include	`#include "imagekit/imagekit.h"`
Link	`lib/imagekit.imp.a`
Library	`SD:/lib/imagekit.so` — 46 entries in its table (`user/Kits/imagekit/imagekit.abi`, append-only)
Sources	`user/Kits/imagekit/`

Using it

ImageKit is the system's one picture codec: BMP, GIF, PNG, JPEG, PCX and WebP read; PNG, JPEG, BMP and GIF written.

A picture read, turned, saved in another format:

```
#include "imagekit/imagekit.h"

ik_image *im = ik_load ("SD:/photos/cat.jpg", 0);
if (im)
{
    int w = ik_width (im), h = ik_height (im);
    unsigned *px = ik_pixels (im);                  // w * h pixels, 0xAARRGGBB
    ik_rotate (im, 1);                              // a quarter turn clockwise
    ik_save (im, "SD:/photos/cat.png", 90);         // the format: the extension's
    ik_image_free (im);
}
```

Pixels of the program's own, saved:

```
ik_image *shot = ik_image_from (pixels, width, height, width);
ik_save (shot, "SD:/screenshots/shot.png", 0);
ik_image_free (shot);
```

The formats, asked — a file chooser's filter follows what the library knows:

```
struct ik_format f[16];
int n = ik_formats (f, 16);
```

Resizing (`ik_scale`), flipping (`ik_flip`), the photo adjustments and the EXIF orientation are in the same header.

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Everything the headers declare, in their order — the details are in each header's part below.

Name	What it does	Header
`ik_free`	a buffer the library returned	`imagekit.h`
`ik_image_new`	transparent black; 0: no memory, a bad size	`imagekit.h`
`ik_image_from`	a copy of pixels (stride: in pixels)	`imagekit.h`
`ik_image_copy`	a new picture with the same pixels; 0: no picture, no memory	`imagekit.h`
`ik_image_free`	the picture and its pixels freed (0: nothing)	`imagekit.h`
`ik_width`	in pixels (0: no picture)	`imagekit.h`
`ik_height`	in pixels (0: no picture)	`imagekit.h`
`ik_format`	IK_ARGB8	`imagekit.h`
`ik_pixels`	its w * h pixels (until the image changes size)	`imagekit.h`
`ik_fill`	every pixel set to that 0xAARRGGBB value	`imagekit.h`
`ik_opaque`	every pixel's alpha 255	`imagekit.h`
`ik_flatten`	laid on a colour, the result opaque	`imagekit.h`
`ik_has_alpha`	1: a pixel is not opaque	`imagekit.h`
`ik_formats`	how many there are (out: up to max)	`imagekit.h`
`ik_info`	(a type)	`imagekit.h`
`ik_is_image_name`	by its extension (bmp gif png jpg jpeg jpe pcx webp)	`imagekit.h`
`ik_probe`	1 / 0: not a picture this	`imagekit.h`

Name	What it does	Header
	reads	
`ik_load`	the picture (an animation: its first frame); 0	`imagekit.h`
`ik_load_mem`	the same from a file's n bytes in memory (flags: ignored, no orientation applied); 0	`imagekit.h`
`ik_load_preview`	the camera's preview, turned as seen; 0: none	`imagekit.h`
`ik_load_format`	the last picture read: "PNG", "JPEG"... ("": none)	`imagekit.h`
`ik_frames_load`	the file read, every frame decoded (a still picture: one frame); 0: not read	`imagekit.h`
`ik_frames_load_mem`	the same from a file's n bytes in memory; 0	`imagekit.h`
`ik_frames_count`	how many frames it holds (0: none)	`imagekit.h`
`ik_frames_pixels`	frame i: w * h pixels (the library's)	`imagekit.h`
`ik_frames_delay`	how long it shows, ms (0: a still picture)	`imagekit.h`
`ik_frames_width`	every frame's width, pixels	`imagekit.h`
`ik_frames_height`	every frame's height, pixels	`imagekit.h`
`ik_frames_free`	the frames it still holds and the handle freed (0: nothing)	`imagekit.h`
`ik_frames_take`	The frames handed over to the caller (px[i]	`imagekit.h`
`ik_inflate`	A deflate stream inflated (zlib	`imagekit.h`
`ik_encode`	0 / -1	`imagekit.h`
`ik_save`	the format from the path's extension -> 0 / -1	`imagekit.h`
`ik_encode_pixels`	Raw pixels encoded (stride	`imagekit.h`
`ik_scale`	The low-level one	`imagekit.h`
`ik_scale_rgb`	The same for pixels whose top byte is not an alpha (0x00RRGGBB, a	`imagekit.h`

Name	What it does	Header
	window's canvas, a photo)	
`ik_resize`	a new picture of that size	`imagekit.h`
`ik_fit`	within the box, proportions kept (grow 0: never larger)	`imagekit.h`
`ik_cover`	the middle cut to that shape and brought to it (a thumbnail)	`imagekit.h`
`ik_crop`	in place -> 0 / -1	`imagekit.h`
`ik_rotate`	clockwise, in place	`imagekit.h`
`ik_flip`	0: left-right, 1: top-bottom	`imagekit.h`
`ik_orient`	1..8 applied	`imagekit.h`
`ik_straighten`	a small angle, enlarged so no corner is empty (the result opaque)	`imagekit.h`
`ik_adjust`	(a type)	`imagekit.h`
`ik_adjust_apply`	in place (the alpha kept)	`imagekit.h`
`ik_adjust_auto`	"enhance": from the picture's histogram	`imagekit.h`
`ik_filter_name`	"Black and white" ...	`imagekit.h`

`imagekit/imagekit.h`

imagekit.h -- ImageKit: pictures, one copy for the whole system (SD:/lib/imagekit.so; the shared libraries: docs/03 section 5.6). A program links lib/imagekit.imp.a and calls plain C functions (ik_*). The first version (2026-10-05) is the base every program that shows or writes a picture needs; the layers, the masks and the brushes of the photo editor come on top of it later.

```
  images      a picture in memory (ik_image): w x h pixels 0xAARRGGBB, straight alpha, rows
one
              after the other
  reading     BMP, GIF (animated too), PNG, JPEG, PCX, WebP -- from a file or from memory;
what a
              file says without decoding it (its size, the camera's orientation, date,
model);
              the camera's own preview
  writing     PNG (zlib's compression, through FileKit), JPEG, BMP, GIF -- to a file or to
memory
  transforms  ONE resize, right in alpha (the average of what a pixel covers when smaller,
              bilinear when larger); fit, cover, crop, quarter turns, flips, the camera's
              orientation applied, straightened by a small angle
  adjustments exposure, contrast, highlights, shadows, saturation, warmth, sharpness and six
              filters as one tone curve; "enhance" from the histogram (Photos' own)
```

Every call takes integers and pointers only: a program built without the FPU calls it. An ik_image is opaque (its format may grow: 16 bits a channel later); a buffer the library returns (void **out) is freed with ik_free.

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```
void ik_free (void *p);             // a buffer the library returned
```

a picture in memory

```
#define IK_ARGB8    1           // 0xAARRGGBB, 8 bits a channel, straight alpha (the only format today)

typedef struct ik_image ik_image;
ik_image *ik_image_new (int w, int h);              // transparent black; 0: no memory, a bad size
ik_image *ik_image_from (const unsigned *px, int w, int h, int stride); // a copy of pixels (stride: in pixels)
ik_image *ik_image_copy (const ik_image *im);   // a new picture with the same pixels; 0: no picture, no memory
void ik_image_free (ik_image *im);  // the picture and its pixels freed (0: nothing)
int ik_width (const ik_image *im);  // in pixels (0: no picture)
int ik_height (const ik_image *im); // in pixels (0: no picture)
int ik_format (const ik_image *im);             // IK_ARGB8
unsigned *ik_pixels (ik_image *im);             // its w * h pixels (until the image changes size)
void ik_fill (ik_image *im, unsigned argb); // every pixel set to that 0xAARRGGBB value
void ik_opaque (ik_image *im);                  // every pixel's alpha 255
void ik_flatten (ik_image *im, unsigned rgb);           // laid on a colour, the result opaque
int ik_has_alpha (const ik_image *im);              // 1: a pixel is not opaque
```

the formats, asked (version 2)

A program lists what the library reads and writes instead of keeping its own list: a format the library learns is one every program has.

```
struct ik_format
{
    char name[12];              // "PNG", "JPEG", "GIF", "BMP", "PCX", "WebP"
    char extensions[40];            // "jpg jpeg jpe" (lower case, a space between)
    int  can_read, can_write;
    int  alpha;             // it keeps transparency
    int  animated;              // it may hold several frames (read as ik_frames)
    int  reserved[4];
};
int ik_formats (struct ik_format *out, int max);        // how many there are (out: up to max)
```

reading

```
#define IK_ORIENT   1           // the camera's orientation (EXIF) applied: as the photo is seen

struct ik_info
{
    int   w, h;             // as stored (before the orientation); 0: not known without decoding
    int   orientation;          // 1..8 (EXIF), 1: as stored
    long long taken;            // when (seconds, local time, from 1970), 0: not known
    char      camera[64];           // "Google Pixel 8" ("": not known)
    char      exposure[64];         // "f/1.7 · 1/640 s · ISO 50 · 6.9 mm"
    int   has_preview;          // the camera's own small picture is in the file (ik_load_preview)
    int   reserved[8];
};
int ik_is_image_name (const char *name);            // by its extension (bmp gif png jpg jpeg jpe pcx webp)
int ik_probe (const char *path, struct ik_info *out);       // 1 / 0: not a picture this reads
ik_image *ik_load (const char *path, int flags);        // the picture (an animation: its first frame); 0
ik_image *ik_load_mem (const void *data, unsigned n, int flags);    // the same from a file's n bytes in memory (flags: ignored, no orientation applied); 0
ik_image *ik_load_preview (const char *path);           // the camera's preview, turned as seen; 0: none
const char *ik_load_format (void);              // the last picture read: "PNG", "JPEG"... ("": none)
```

An animation (a GIF): every frame.

```
typedef struct ik_frames ik_frames;
ik_frames *ik_frames_load (const char *path);   // the file read, every frame decoded (a still picture: one frame); 0: not read
ik_frames *ik_frames_load_mem (const void *data, unsigned n);   // the same from a file's n bytes in memory; 0
int ik_frames_count (const ik_frames *f);   // how many frames it holds (0: none)
const unsigned *ik_frames_pixels (const ik_frames *f, int i);   // frame i: w * h pixels (the library's)
int ik_frames_delay (const ik_frames *f, int i);        // how long it shows, ms (0: a still picture)
int ik_frames_width (const ik_frames *f);   // every frame's width, pixels
int ik_frames_height (const ik_frames *f);  // every frame's height, pixels
void ik_frames_free (ik_frames *f); // the frames it still holds and the handle freed (0: nothing)
```

The frames handed over to the caller (px[i]: new unsigned[w * h], the caller's to delete []; delay[i]: ms): up to max of them -> how many; f keeps none of those. (UIKit's img_load is this.)

```
int ik_frames_take (ik_frames *f, unsigned **px, int *delay, int max);
```

A deflate stream inflated (zlib: with its header, as PNG's; else raw): a new unsigned char[] of *out_n bytes (the caller's to delete []), or 0. (UIKit's img_inflate; a new program takes FileKit's.)

```
unsigned char *ik_inflate (const void *data, unsigned n, int zlib, unsigned *out_n);
```

writing

format: "png", "jpg" / "jpeg", "bmp", "gif" (letters' case ignored). quality: JPEG's 1..100 (0: 90). A picture with transparency keeps it in PNG; GIF keeps its clear pixels (alpha < 128) clear and the others opaque; JPEG and BMP lay it on white.

```
int ik_encode (const ik_image *im, const char *format, int quality, void **out, unsigned
*out_n);   // 0 / -1
int ik_save (const ik_image *im, const char *path, int quality);    // the format from the
path's extension -> 0 / -1
```

Raw pixels encoded (stride: pixels a row). alpha: 1 kept where the format has it (PNG: RGBA), 0 not (PNG: RGB, whatever the top bytes hold), -1 kept when a pixel is not opaque. (img/pngsave.hpp's encoders are this, in the programs built with PNGSAVE_USE_IMAGEKIT.)

```
int ik_encode_pixels (const unsigned *px, int w, int h, int stride, const char *format, int
quality, int alpha,
              void **out, unsigned *out_n);
```

transforms

The low-level one: the rectangle (sx, sy, sw, sh) of src (src_w x src_h, src_stride pixels a row) brought to dst (dw x dh, dst_stride pixels a row). Right in alpha (the colours weighed by it). A big shrink reads at most ~8 x 8 of the pixels each result covers (a 24 Mpx photo's thumbnail stays quick).

```
void ik_scale (const unsigned *src, int src_w, int src_h, int src_stride, int sx, int sy, int
sw, int sh,
           unsigned *dst, int dst_stride, int dw, int dh);
```

The same for pixels whose top byte is not an alpha (0x00RRGGBB, a window's canvas, a photo): the colours alone; the result's top byte is 0.

```
void ik_scale_rgb (const unsigned *src, int src_w, int src_h, int src_stride, int sx, int sy,
int sw, int sh,
           unsigned *dst, int dst_stride, int dw, int dh);
ik_image *ik_resize (const ik_image *im, int w, int h);     // a new picture of that size
ik_image *ik_fit (const ik_image *im, int max_w, int max_h, int grow);  // within the box,
proportions kept (grow 0: never larger)
ik_image *ik_cover (const ik_image *im, int w, int h);      // the middle cut to that shape
and brought to it (a thumbnail)
int ik_crop (ik_image *im, int x, int y, int w, int h);     // in place -> 0 / -1
int ik_rotate (ik_image *im, int quarter_turns);        // clockwise, in place
int ik_flip (ik_image *im, int vertical);           // 0: left-right, 1: top-bottom
int ik_orient (ik_image *im, int exif_orientation);     // 1..8 applied
int ik_straighten (ik_image *im, int millidegrees);     // a small angle, enlarged so no
corner is empty (the result opaque)
```

adjustments (Photos')

```
#define IK_FILTER_NONE      0
#define IK_FILTER_BW        1
#define IK_FILTER_WARM      2
#define IK_FILTER_COOL      3
#define IK_FILTER_VINTAGE   4
#define IK_FILTER_VIVID     5
#define IK_FILTERS      6

struct ik_adjust
```

```
{
    int exposure, contrast, highlights, shadows, saturation, warmth;    // -100..100, 0: as
it is
    int sharpness;                                  // 0..100
    int filter;                                 // IK_FILTER_*
    int reserved[8];                                // 0
};
void ik_adjust_apply (ik_image *im, const struct ik_adjust *a); // in place (the alpha kept)
void ik_adjust_auto (const ik_image *im, struct ik_adjust *out);    // "enhance": from the
picture's histogram
const char *ik_filter_name (int filter);            // "Black and white" ...

}
#endif
```